

## SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

---

### COURSE OUTCOME COVERED

**CO1:** Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

---

### Short Answer Test

91.5%

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every

window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.
4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.
6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.



1. Given:

File size = 150 MB

= 150,000,000 bytes

Bandwidth = 50 Mbps

segment size = 1500 bytes

Data link overhead = 40 bytes / frame

Calculations:

Step -1: Number of segments

$$\text{Segments} = \frac{150,000,000}{1500} = 100,000$$

Each segment is placed into one frame. Therefore, the number of frames is equal to the number of segments.

Step 2: Number of frames

$$\text{Frames} = 100,000$$

The data link layer adds 40 bytes of overhead to every frame. This overhead contains control information such as addressing and error checking.



Step-3: calculate total overhead:

$$100,000 \times 40 = 4,000,000 \text{ bytes}$$
$$= 4 \text{ MB}$$

Step-4: calculate total data transmitted

$$150 + 4 = 154 \text{ MB}$$

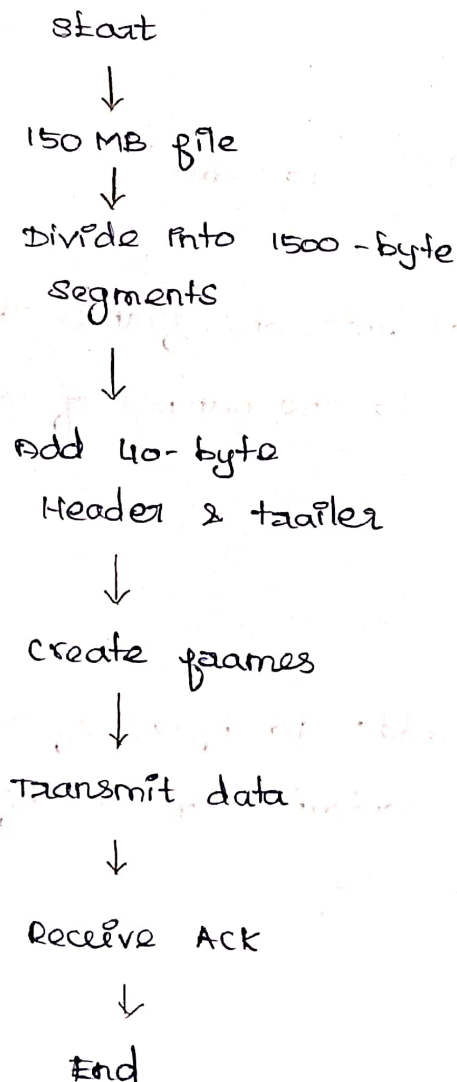
Step-5: calculate transmission Time

convert MB to mb

$$154 \times 8 = 1232 \text{ mb}$$

$$\text{Transmission time} = \frac{1232}{50} = 24.64 \text{ sec}$$

Flow chart:



## Reliable communication:

### transport layer:

Ensures reliable communication through segmentation, sequence numbers, acknowledgements and retransmission of lost segments.

### Data link layer:

Ensures reliable frame delivery using headers, trailers, CRC based error detection and retransmission of corrupted frames.

2.

The transport layer uses the sliding window protocol to improve communication efficiency. A sliding window allows the sender to transmit multiple frames before waiting for an acknowledgement. This reduces waiting time and improves network throughput.

#### Given:

sliding window size = 6 frames

User data per frame = 1024 bytes

total frames = 48

Step 1: calculate the number of transmission window:

$$\begin{aligned} \text{Transmission} \\ \text{window} &= \frac{48}{6} = 8 \end{aligned}$$

Assume one frame is lost in every window. Each lost frame must be retransmitted.

step 2: calculate total retransmissions:

no of windows = 8

lost frames = 1

Retransmissions =  $8 \times 1 = 8$

Flow control:

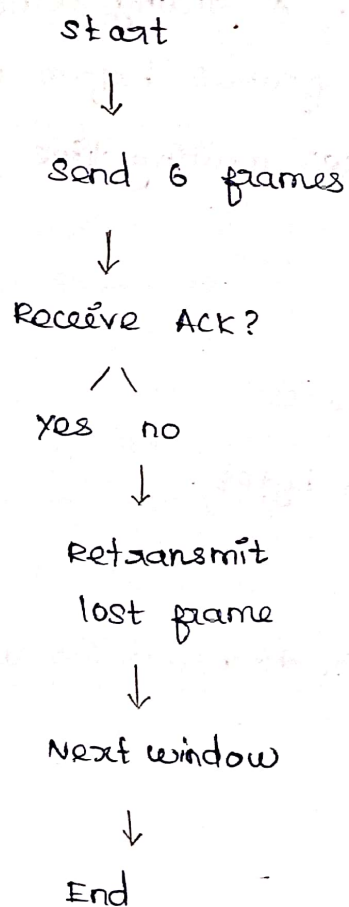
Flow control prevents the sender from transmitting data faster than the receiver can process it.

Prevents buffer overflow at receiver

Reduces data loss and retransmissions

Improves communication efficiency and throughput.

Flow chart:





Sender

Receiver

|F<sub>1</sub>| |F<sub>2</sub>| |F<sub>3</sub>| |F<sub>4</sub>| |F<sub>5</sub>| |F<sub>6</sub>|

| | | | |

→

← X F<sub>4</sub> lost

Ack for received frames

Retransmit F<sub>4</sub>

|

-----→

received

### 3. Theory:

The network layer is responsible for routing packets from the source to the destination. If a packet is larger than the MTU (Maximum Transmission Unit) it is divided into smaller fragments. Each fragment carries its own 20 byte IP header.

#### Given:

IP packet size = 12000 bytes

MTU = 1500 bytes

IP header = 20 bytes

#### Step 1: calculate data per fragment

Each fragment carry 1480 bytes of data because 20 bytes are used by the IP header.

$$1500 - 20 = 1480$$

Step -2: calculate number of fragments:

$$\frac{18000}{1480} = 8.11$$
$$= 9$$

Step -3: calculate total header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Network layer reliability:

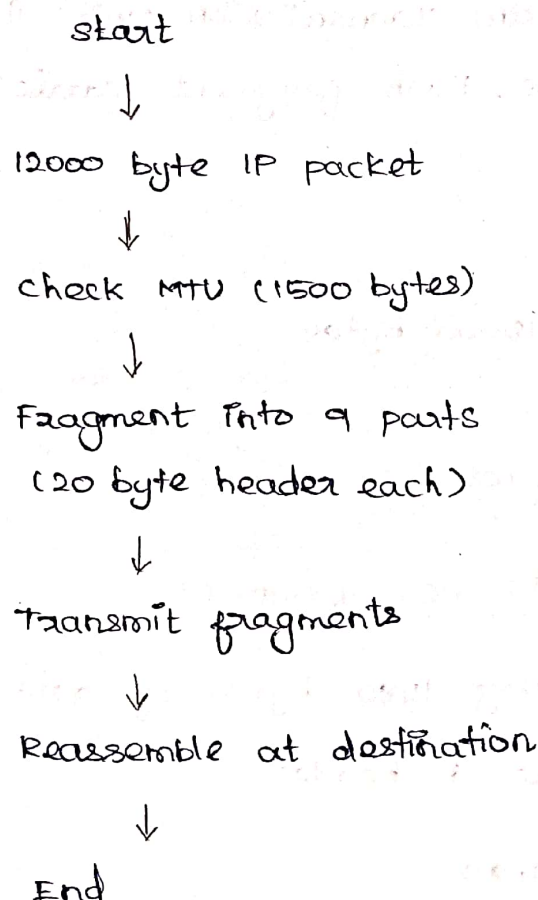
It divides large packets into smaller fragments

Assigns a fragments offset and identification fields

Routes each fragment independently

The destination reassembles all fragments into the original packet using the fragment information.

Flow chart:



original packet (12000 bytes)



| Fragmentation |



| F<sub>1</sub> | | F<sub>2</sub> | | F<sub>3</sub> | | F<sub>4</sub> |



Destination (reassembles packet)

#### 4. Theory:

The transport layer uses the sliding window protocol to send multiple frames before receiving an acknowledgement. This improves data transmission speed and supports flow control, preventing the sender from overwhelming the receiver.

#### Given:

sliding window size = 6 frames

Data per frame = 1024 bytes

Total frames = 48

Step 1: calculate number of transmission windows:

$$\text{no of windows} = \frac{\text{total frames}}{\text{sliding window size}}$$

$$= \frac{48}{6} = 8$$

If one frame is lost in every window, that lost frame must be retransmitted.

Step 2: calculate total retransmissions.

Transmission window  $\times$  lost frames

$$= 8 \times 1 = 8$$

Flow control:

Flow control improves communication efficiency by

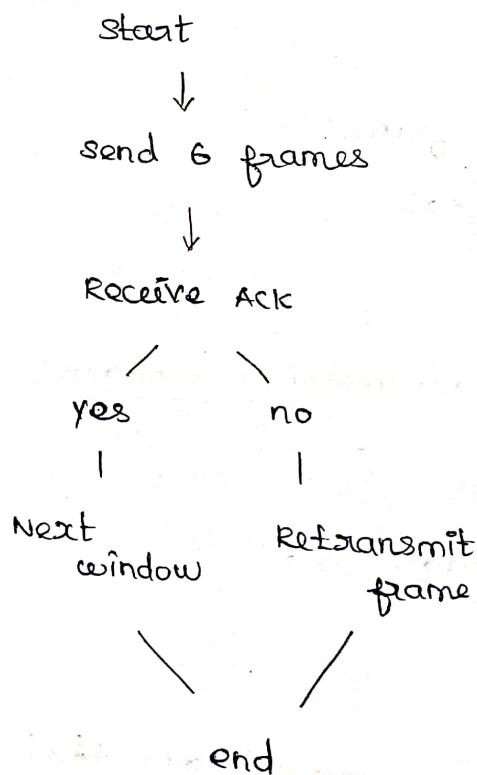
Preventing the sender from sending data faster than the receiver can process.

Avoiding receiver buffer overflow

Reducing data loss and unnecessary retransmissions

Improving network throughput and reliability.

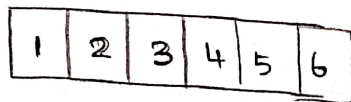
Flow chart:



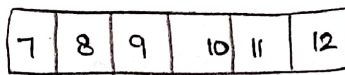


Sender

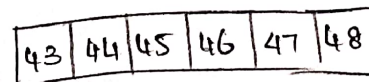
Frames



window 1

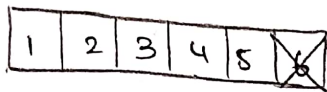


window 2

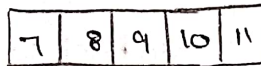


window 8

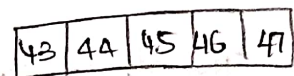
Receiver



frame lost



frame lost



frame lost

ACK 1-5

ACK 7-11

ACK 43-47

5. total file size = 600 MB

checkpoint after every 60 MB

Failure occurs after 390 MB

calculations:

1) Number of checkpoints created before failure

checkpoints are inserted every 60 MB

Failure occurs after 390 MB

No of checkpoints =  $\frac{390}{60} = 6.5 \Rightarrow 6$  checkpoints

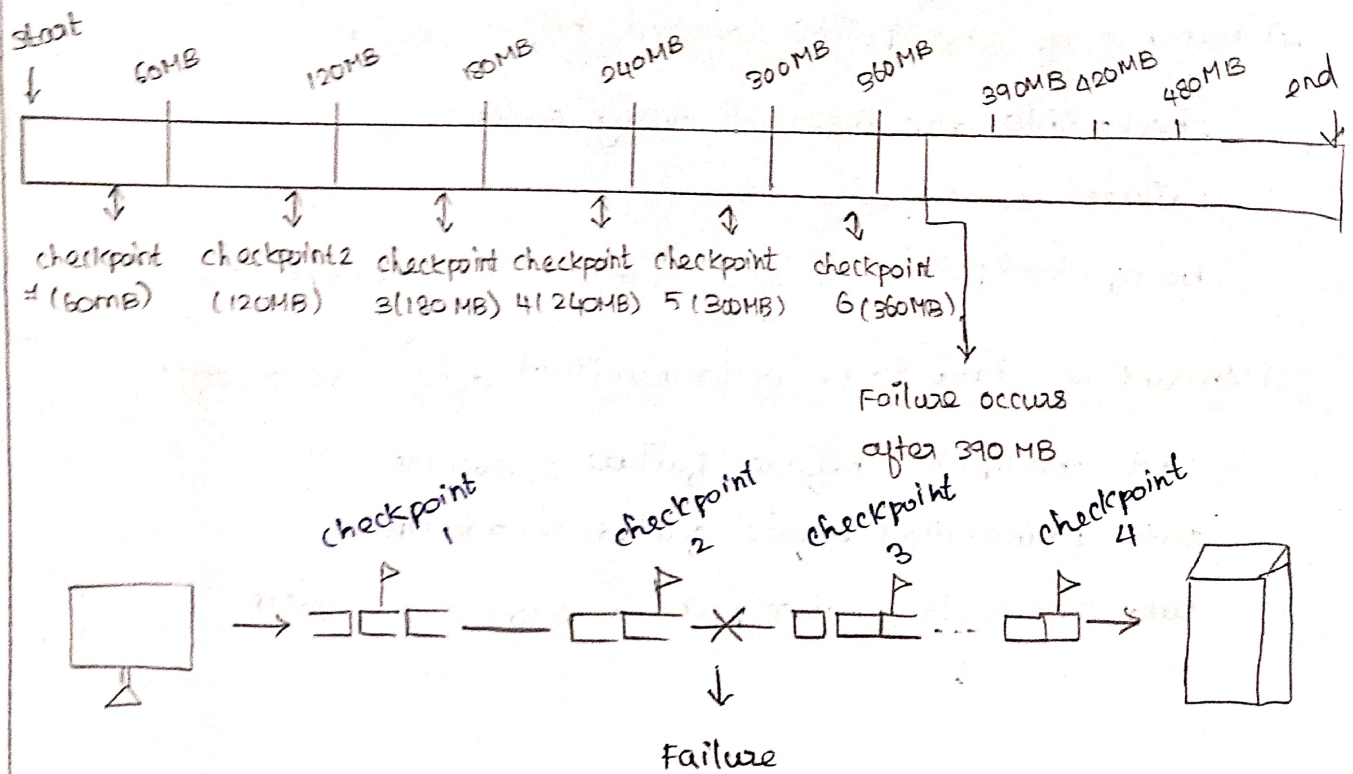
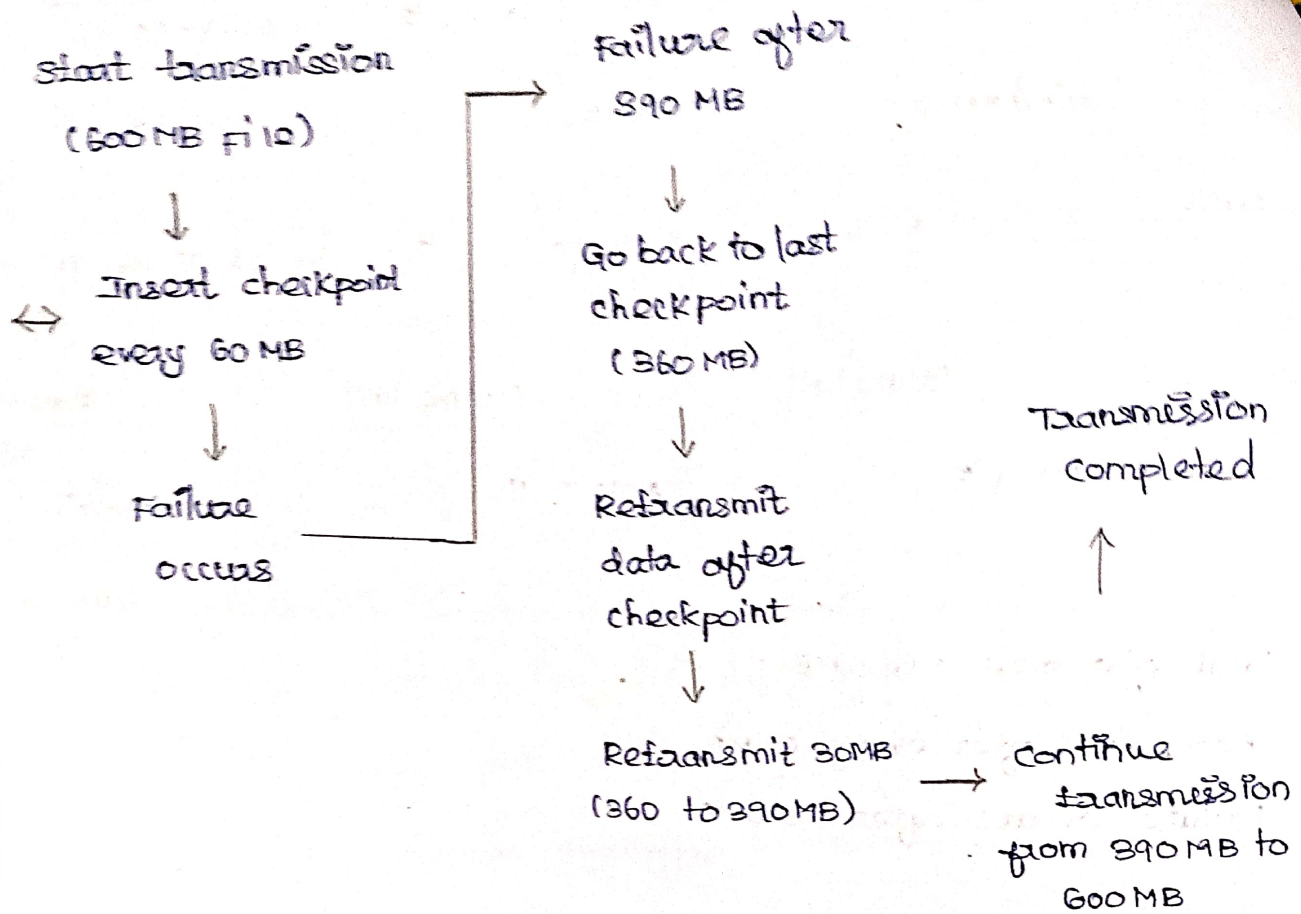
2) Amount of data to be retransmitted after recovery:

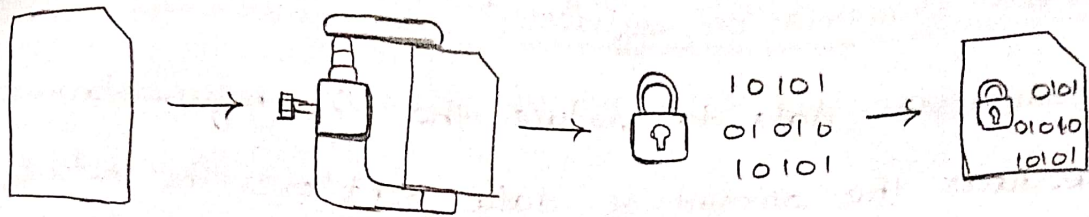
Last checkpoint before failure = 360 MB

data transmitted before failure = 390 MB

data after last checkpoint =  $390 - 360 = 30$  MB

# Transmission & Recovery flow:





original  
file  
900 MB.

Compression  
reduces size  
by 40%.

Encryption  
adds 5%  
overhead

Encrypted file  
(ready for  
transmission)

### calculations:

#### 1. Compressed file size:

$$\begin{aligned}\text{Compression} &= 40\% \text{ of } 900 \text{ MB} \\ &= 0.40 \times 900 \\ &= 360 \text{ MB}\end{aligned}$$

$$\text{Compressed size} = 900 - 360 = 540 \text{ MB}$$

#### 2. Encrypted file size:

$$\begin{aligned}\text{Encryption overhead} &= 5\% \text{ of compressed size} \\ &= 0.05 \times 540 \\ &= 27 \text{ MB}\end{aligned}$$

$$\text{Encrypted size} = 540 + 27 = 567 \text{ MB}$$

#### 3. Total percentage reduction:

$$\begin{aligned}\text{Reduction} &= \text{original} - \text{encrypted} \\ &= 900 - 567 \\ &= 333 \text{ MB}\end{aligned}$$

$$\begin{aligned}\text{Percentage reduction} &= (333/900) \times 100 \\ &= 37\%\end{aligned}$$



## Improves transmission efficiency:

compresses data to reduce the size before transmission  
Reduces the amount of data sent over the network  
Saves bandwidth and storage  
Results in faster transmission and lower cost.

## Improves security:

Encrypts the compressed data  
Protects data from unauthorized access  
Ensures confidentiality, integrity and authenticity  
Prevents snooping and data tampering.

7.

### Problem scenario:

Total data to transfer = 3GB

Effective throughput = 120 Mbps

Application layer request size = 3MB

Protocol = FTP (Application layer)

### 1. Total number of requests:

request size = 3MB

total data = 3GB

$$\text{no of requests} = \frac{\text{request total data}}{\text{request size}} = 3$$

$$1 \text{ GB} = 1024 \text{ MB} \Rightarrow 3 \text{ GB} = 3 \times 1024$$

$$= 3072 \text{ MB}$$

$$= \frac{3072}{3} = 1024 \text{ requests}$$



total transmission time:

total data = 3GB = 3072 MB

1 byte = 8 bits

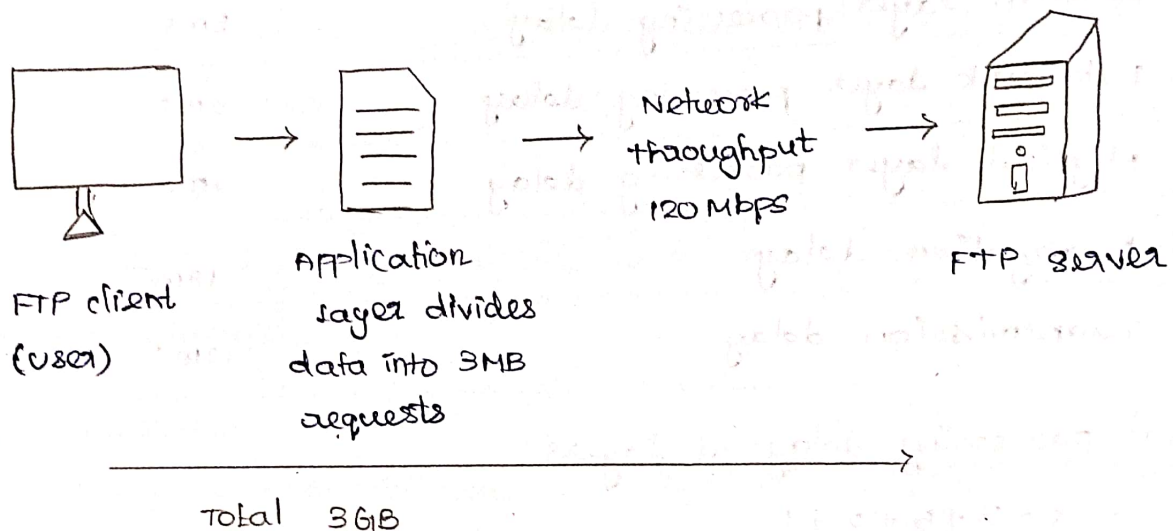
$$\begin{aligned}\text{Total data in bits} &= 3072 \times 1024 \times 1024 \times 8 \\ &= 25,169,803,776 \text{ bits}\end{aligned}$$

$$\text{throughput} = 120 \text{ Mbps} = 120 \times 10^6 \text{ bits/sec}$$

$$\text{transmission time} = \frac{\text{total data in bits}}{\text{throughput}}$$

$$= \frac{25,169,803,776}{120 \times 10^6}$$

$$= 214.75 \text{ sec.}$$



User interface & interaction:

Provides an easy interface for users to request and access remote files.

File management:

Supports operations like upload, download, delete, rename, and list files.

## Reliability:

uses Acknowledgement and error messages to ensure successful transfer and informs users of any failures

## Data Integrity:

Ensures that the complete and correct data is delivered to the user.

8.

| layer / delay type                 | delay |
|------------------------------------|-------|
| Application layer processing delay | 4ms   |
| Transport layer processing delay   | 3ms   |
| Network layer processing delay     | 5ms   |
| Data link layer processing delay   | 2ms   |
| Physical layer processing delay    | 1ms   |
| Propagation delay                  | 18ms  |
| Transmission delay                 | 12ms  |

### Total processing delay at layers:

$$= 4 + 3 + 5 + 2 + 1$$

$$= 15\text{ms}$$

### Total end to end delay

$$= \text{Total processing delays} - \text{Transmission delay} + \text{propagation delay}$$

$$= 15\text{ms} + 12\text{ms} + 18\text{ms}$$

$$= 45\text{ms}$$

### Reliability:

uses Acknowledgement and error messages to ensure successful transfer and informs users of any failures

### Data Integrity:

Ensures that the complete and correct data is delivered to the user.

8.

| layer / delay type                 | delay |
|------------------------------------|-------|
| Application layer processing delay | 4ms   |
| Transport layer processing delay   | 8ms   |
| Network layer processing delay     | 5ms   |
| Data link layer processing delay   | 2ms   |
| Physical layer processing delay    | 1ms   |
| Propagation delay                  | 18ms  |
| Transmission delay                 | 12ms  |

### Total processing delay at layers:

$$= 4 + 3 + 5 + 2 + 1$$

$$= 15\text{ms}$$

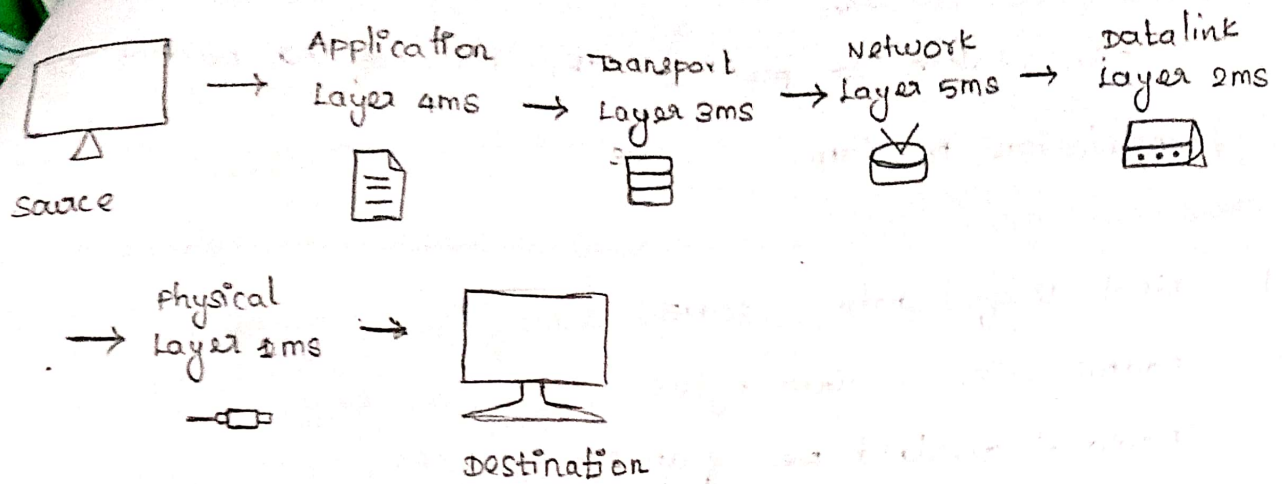
### Total end to end delay

$$= \text{Total processing delays} - \text{transmission delay} + \text{propagation delay}$$

$$= 15\text{ms} + 12\text{ms} + 18\text{ms}$$

$$= 45\text{ms}$$





### Application layer:

Provides services to user applications (e.g: HTTP, FTP)  
prepares data for communications.

### Transport layer:

Ensures end to end delivery, segmentation, flow control, and error control (e.g: TCP, UDP)

### Network layer:

Handles logical addressing and routing of packets across multiple networks.

### Datalink layer:

Provides node to node delivery, framing, MAC addressing, and error detection.

### Physical layer:

Transmits raw bits over the physical medium (e.g: cables, signals)

### Propagation delay:

Time taken for the signal to travel across the physical medium.



### Transmission delay:

Time taken to push all the bits of the packet onto transmission medium.

9.

Total Useful data = 80 MB

Frame size = 1600 bytes

Protocol overhead per frames = 80 bytes

Useful data per frame =  $1600 - 80 = 1520$  bytes

80 MB =  $80 \times 1024 \times 1024$

= 83,886,080 bytes

### Frame structure:

Each frame of 1600 bytes contains Useful data + protocol overhead.

Useful data  
1520 bytes

protocol overhead  
80 bytes

overhead added by different OSI layers

|             |           |          |           |          |
|-------------|-----------|----------|-----------|----------|
| Application | Transport | Network  | data link | physical |
| 20 bytes    | 20 bytes  | 20 bytes | 10 bytes  | 10 bytes |

Total overhead = 80 bytes

### calculations:

4. Total number of frames required:

Useful data per frame = 1520 bytes

Total Useful data = 83,886,080

no of frames =  $\frac{\text{total Useful data}}{\text{Useful data per frame}}$



$$= \frac{83,886,080}{1520} = 55,190$$

2. total overhead frames:

overhead per frame = 80 bytes

total overhead = no of frames  $\times$  overhead per frame

$$= 55,190 \times 80$$

$$= 4,415,200 \text{ bytes}$$

$$= 4208 \text{ MB (approx)}$$

3. Transmission efficiency:

$$\text{Efficiency} = \frac{\text{Useful data}}{\text{total data transmitted}} \times 100$$

$$= \frac{\text{Useful data} + \text{overhead}}{\text{total data transmitted}}$$

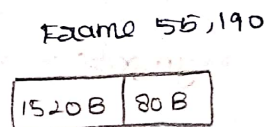
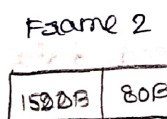
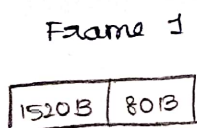
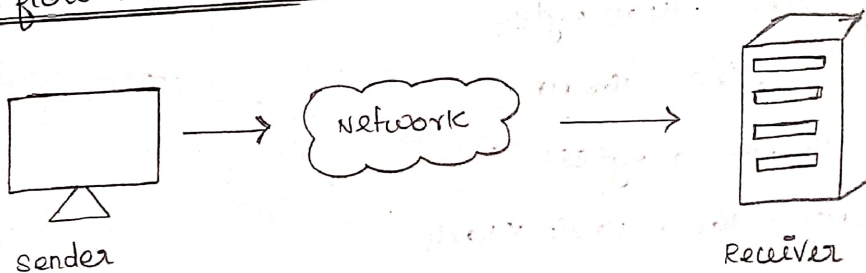
$$= 83,886,080 + 4,415,200$$

$$= 88,301,280 \text{ bytes}$$

$$= \frac{83,886,080}{88,301,280} \times 100$$

$$= 94.99\% \text{ (approx)}$$

Data flow illustration:



$$\begin{aligned}
 \text{Total data transmitted} &= \text{No of frames} \times \text{frame size} \\
 &= 55,190 \times 1600 \\
 &= 88,301,280 \text{ bytes}
 \end{aligned}$$

10.

| Parameter                         | Value      |
|-----------------------------------|------------|
| original medical record size      | 240 MB     |
| compression by presentation layer | 30%        |
| segment size (transport layer)    | 1400 bytes |
| overhead per frame                | 60 bytes   |

calculations:

1. compressed file size (presentation layer)

$$\text{compression} = 30\%$$

$$\begin{aligned}
 \text{compressed size} &= \text{original size} - (100 - 30\%) \\
 &= 240 \text{ MB} - 70\% \\
 &= 168 \text{ MB}
 \end{aligned}$$

2. total number of segments (transport layer)

$$\text{segment size} = 1400 \text{ bytes}$$

$$\text{compressed size} = 168 \text{ MB}$$

convert 168 MB to bytes:

$$\begin{aligned}
 168 \text{ MB} &= 168 \times 1024 \times 1024 \\
 &= 176,160,768
 \end{aligned}$$

$$\text{no of segments} = \frac{\text{total compressed data}}{\text{segment size}}$$

$$= \frac{176,160,768}{1400} = 125,830$$





## protocol overhead

overhead per frame = 60 bytes

total overhead = no of frames  $\times$  overhead per frame

$$= 125830 \times 60$$

$$= 7549800 \text{ bytes}$$

$$= 7.20 \text{ MB}$$

### 4. Final amount of data transmitted across physical layer

Final data = compressed data + total overhead

$$= 176,160,768 + 7,549,800$$

$$= 183,710,568 \text{ bytes}$$

$$= 175.1 \text{ MB}$$

## Data processing through OSI layers:

### Application Layer:

The hospital application requests to send the patient medical record.

### Presentation layer:

compress the data by 30% to reduce size and may also encrypt for security.

### Transport Layer:

Divides the compressed data into 1400 bytes segments and executes reliable end to end delivery.

### Network layer:

Adds logical addressing and executes the packets to the destination.



## Data link layer:

creates frames and adds 60 bytes of overhead to each segment.

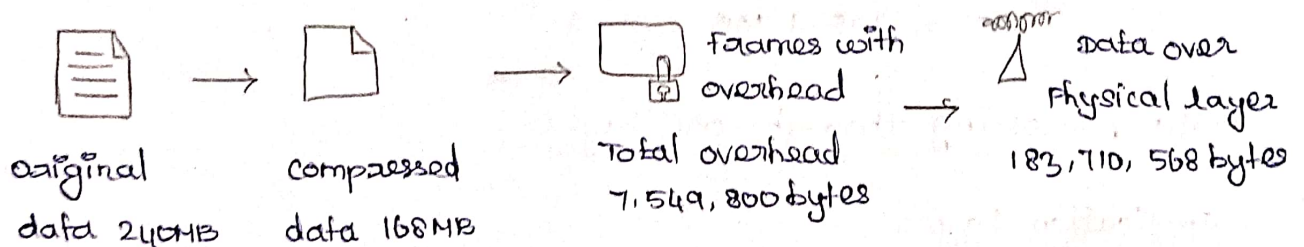
## Physical layer:

converts frames into bits and transmits them over the physical medium.

Final data transmitted

= 183,710,568 bytes

= (175.1 MB)



|                      | Q1            | Q2               | Q3               | Q4            | Q5            | Q6            | Q7            | Q8               | Q9            | Q10           |
|----------------------|---------------|------------------|------------------|---------------|---------------|---------------|---------------|------------------|---------------|---------------|
|                      | L3L2L1        | L3L2L1           | L3L2L1           | L3L2L1        | L3L2L1        | L3L2L1        | L3L2L1        | L3L2L1           | L3L2L1        | L3L2L1        |
| or descending<br>45% | 2.25          | 2                | 2.25             | 2             | 2             | 2.25          | 2.25          | 2                | 2.25          | 2             |
| Explanation<br>35%   | 1.75          | 1.75             | 1                | 1             | 1             | 1.75          | 1.75          | 1.75             | 1.75          | 1             |
| clarity<br>20%       | $\frac{1}{5}$ | $\frac{1}{4.75}$ | $\frac{1}{4.25}$ | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{1}{5}$ | $\frac{1}{5}$ | $\frac{1}{4.75}$ | $\frac{1}{5}$ | $\frac{1}{4}$ |

~~AB~~ A5.75